

Question Set 5

Comments: some questions in the question set are not like exam or test questions. They are used for the purpose of reflecting on some topic to ensure students understand the main concepts. These will be called here "Reflecting questions" (RQ). The questions more similar to exam/test questions will be labelled ETQ.

In class questions:

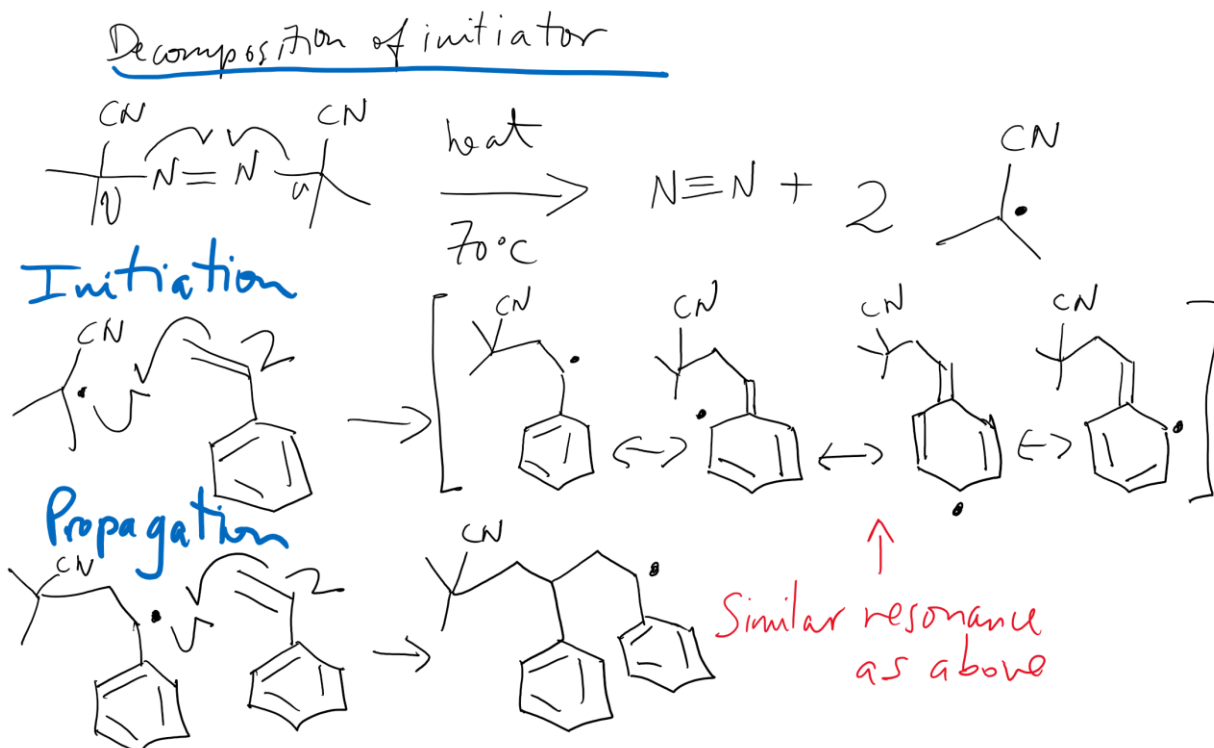
- 1) Have a look at the Lecture Notes Set 3 (Slides 1-37 only) and ask questions of what you do not understand.
- 2) When carrying out free-radical polymerisation you will find that monomers and solvents are very well-purified taking out anything that it is not monomer. In addition, reactions are carried out in inert atmosphere (N_2 or Ar gas atmosphere), solvents and reactants are "degassed" via bubbling through a stream of N_2 gas or through procedures called "freeze, pump, thaw". Explain why this is necessary.

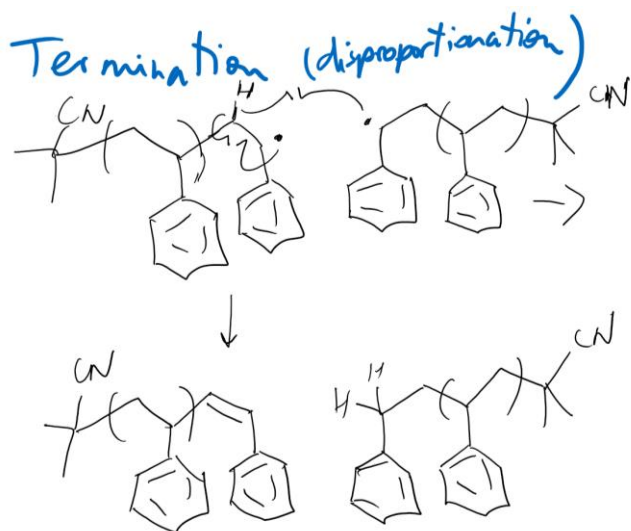
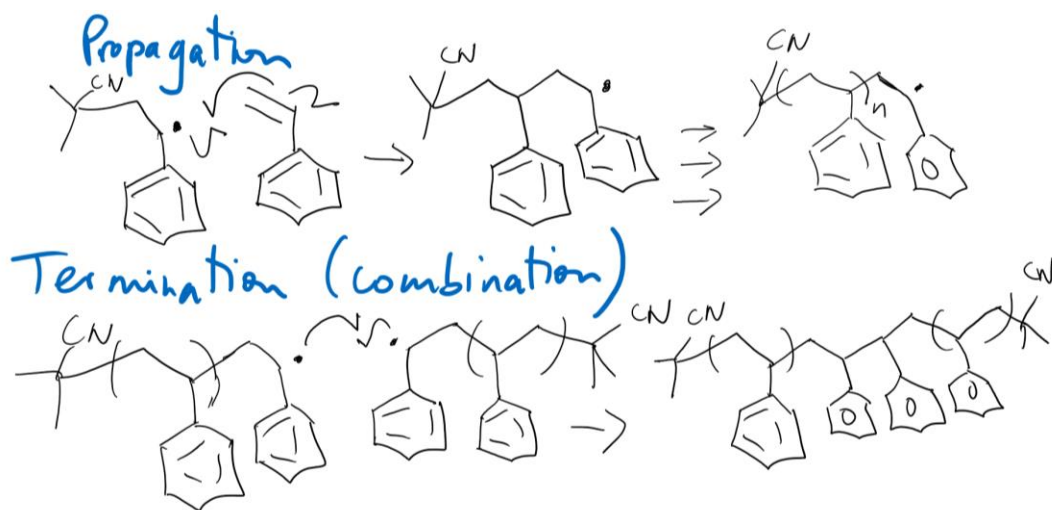
RQ/ETQ: Impurities can have an impact on free-radical polymerisations. In particular, monomers are sold containing an inhibitor (usually a chain-transfer agent) that prevents polymerisation inside the monomer bottle. These react with free-radicals. Another undesired "impurity" is oxygen gas, present in all liquids, as well as in normal air. Oxygen also reacts with free-radicals.

If inhibitors/oxygen are not removed free-radical initiators are wasted reacting with the inhibitors/oxygen, until the latter have been depleted. This would cause a "lag period" during which polymerisation cannot occur. It also makes difficult to calculate the amount of initiator needed for polymerisation, because you can never be too sure how much oxygen/inhibitors precisely there is in a monomer sample.

- 3) Styrene is polymerised free-radically at 70 °C using AIBN as initiator. Draw the curly arrow mechanism of the polymerisation clearly indicating initiator degradation, initiation, propagation and termination.

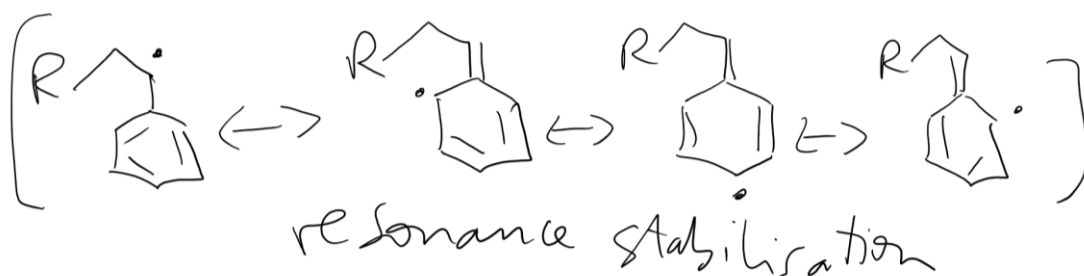
ETQ





4) Why is styrene such a good monomer for free-radical polymerisation? (Use structures in your answer)

ETQ: Free-radicals are not stable species. Some level of free-radical stabilisation sometimes helps a monomer polymerise by FRP. Styrene forms a benzylic free-radical stabilised by resonance as shown below.



5) Why increasing initiator concentration reduces molecular weight in free-radical polymerisation?

ETQ: Increasing initiator concentration will increase the concentration of free-radicals in a polymerisation. This will increase the rate of all the processes involved (decomposition of initiator, initiation, propagation and termination). It can be seen from the rate equations below why this is the case:

$$v_i = \frac{d[M\bullet]}{dt} = 2k_d f[i]$$

$$v_p = k_p [M][M\bullet]$$

$$v_t = 2k_t[M\bullet][M\bullet] = 2k_t[M\bullet]^2$$

However, these processes, are not equally affected by concentration of free-radicals (hence of initiator). While initiation is first-order with respect to [i] and propagation is first-order with respect to free radical concentration, termination is second-order with respect to free-radical concentration. SO, increasing [i] increases the rate of termination much more than initiation or propagation, making polymers smaller. This can also be seen in the equation of average kinetic chain length, where [i] is part of the denominator.

$$\bar{v} = \frac{k_p[M]}{2\sqrt{k_d k_t f[i]}}$$

- 6) **ETQ:** Suppose two free-radical polymerisation reactions in parallel. The first one has styrene and the second one vinyl acetate. Write the structure of each polymer. Given the data and supposing equal conditions (monomer and initiator concentration, temperature and initiator efficiency) calculate the ratio in average chain length between the two polymers.

$$\bar{v} = \frac{k_p[M]}{2\sqrt{k_d k_t f[i]}}$$

$$\frac{\bar{v}_{VA}}{\bar{v}_{ST}} = \frac{k_{pVA} [VA] \cancel{2\sqrt{k_d k_{tST} f_{ST}[i]}}}{\cancel{2\sqrt{k_d k_{tVA} f_{VA}[i]}} k_{pST} [ST]} = \frac{k_{pVA} \sqrt{k_{tST}}}{\sqrt{k_{tVA}} k_{pST}}$$

$$[VA] = [ST] \quad f_{VA} = f_{ST} \quad \Rightarrow \quad \frac{\bar{v}_{VA}}{\bar{v}_{ST}} = 11.8$$

What does this mean for the molecular weight of both polymers prepared under these conditions?

It means it is expected that Polyvinyl acetate will be longer. Of course, we are not considering chain-transfer, therefore, we cannot affirm this for sure. ALSO, the equations used only apply during steady-state kinetic period of the polymerisation: we cannot anticipate the relative length of polymer formed earlier or later.

Give reasons for your answer using equations. How would you change the reaction conditions so that both polymers grow to the same length?

Use a different concentration of monomer:

$$\bar{v} = \frac{k_p [M]}{2 \sqrt{k_d k_t f [i]}}$$

$\bar{v}_{VA} = \bar{v}_{ST}$

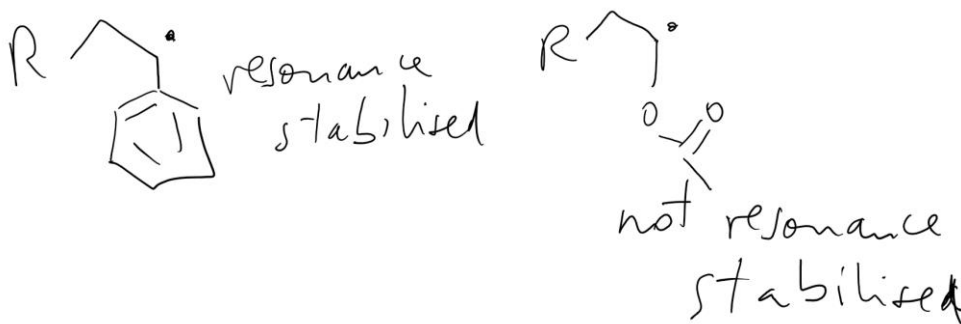
aim

$$\frac{k_{pVA} [VA]}{2 \sqrt{k_d k_{tVA} f_{VA} [i]}} = \frac{k_{pST} [ST]}{2 \sqrt{k_d k_{tST} f_{ST} [i]}}$$

$$\Rightarrow \frac{k_{pVA} \sqrt{k_{tST}}}{\sqrt{k_{tVA}} k_{pST}} = \frac{[ST]}{[VA]} = 11.8 \rightarrow \text{to "reverse" the effect of } k$$

Provide an explanation using chemical structures for the differences in the corresponding kinetic constants.

The fact k for styrene are smaller than corresponding k for vinyl acetate, means that the activation energies are higher. The activation energies relate to the difference in energy between starting species in the reaction and the energy of the transition states. It is expected that the energy of transition states are similar. However, the starting species are free-radicals, which are more stable for styrene than for vinyl acetate, because of the resonance stabilisation of the benzylic FR and no resonance for the VA FR (see Question 4 and below).



This is like stating that the styrene free radical is more stable, therefore, less reactive, than the VA free-radical.

Styrene: $k_p = 176 \text{ L/mol s}$; $k_t = 72 \times 10^6 \text{ l/mol s}$

Vinyl acetate: $k_p = 2640 \text{ L/mol s}$; $k_t = 117 \times 10^6 \text{ l/mol s}$

